Δ vs frozen (good-oriented, value + 95% CI) · PRETRAIN block / SURGERY block · SCORE = mean per-metric-normalized (0–1) · BLUE BOX + big Δ = best arm per metric
WINNER row (champion duel — S=surgery, P=pretrain, ==tie): surgery 1 · pretrain 8 · tie 0 → PRETRAIN

	action_top1 ↑ better classify motion class from frozen features	motion_cos ↑ better same-motion clips cluster vs different motions	taxonomy_f1 ↑ better read scene attributes from features	future_mse ↓ better predictor guesses masked next-frame tokens	rollout ↓ better error growth over multi-step rollout	causal ↓ better predict 2nd half from 1st half	tdist ↓ better accuracy decay vs prediction horizon	teacher_free ↓ better free-run minus teacher-forced gap	maskratio ↓ better degradation under heavier masking	SCORE (norm 0–1)
ViT-g·2.0 pretrain_2X encoder	+11.62 [+9.260, +13.918]	**+0.221** [+0.215, +0.226]	-0.019 [-0.037, -0.002]	-0.138 [-0.142, -0.135]	-0.0005 [-0.001, -0.000]	-0.120 [-0.123, -0.116]	-0.0009 [-0.001, -0.000]	-0.0089 [-0.009, -0.009]	-0.018 [-0.019, -0.018]	0.31
ViT-g·2.0 pretrain encoder	+11.67 [+9.479, +13.918]	+0.105 [+0.102, +0.109]	-0.0084 [-0.026, +0.009]	-0.110 [-0.113, -0.107]	**+0.0030** [+0.003, +0.003]	-0.088 [-0.090, -0.085]	**+0.0031** [+0.003, +0.004]	-0.0019 [-0.002, -0.002]	-0.010 [-0.011, -0.010]	0.65
ViT-g·2.0 pretrain head	+0.603 [-1.589, +2.740]	+0.0018 [+0.002, +0.002]	**+0.0005** [-0.015, +0.016]	**-0.0003** [-0.001, +0.000]	0 [+nan, +nan]	**0** [+nan, +nan]	0 [+nan, +nan]	**0** [+nan, +nan]	**0** [+nan, +nan]	0.62
ViT-g·2.0 surgery 3stage_DI encoder	+10.90 [+8.658, +13.206]	+0.113 [+0.110, +0.117]	-0.0075 [-0.025, +0.010]	-0.073 [-0.076, -0.070]	-0.0017 [-0.002, -0.001]	-0.065 [-0.068, -0.062]	+0.0010 [+0.001, +0.001]	-0.019 [-0.019, -0.018]	-0.0078 [-0.009, -0.007]	0.44
ViT-g·2.0 surgery 3stage_DI head	**+11.89** [+9.699, +14.192]	+0.112 [+0.108, +0.115]	-0.0079 [-0.025, +0.009]	-0.110 [-0.112, -0.107]						0.57
ViT-g·2.0 surgery noDI encoder	+11.29 [+8.986, +13.589]	+0.116 [+0.113, +0.120]	-0.013 [-0.030, +0.004]	-0.070 [-0.072, -0.067]	-0.0008 [-0.001, -0.000]	-0.062 [-0.065, -0.060]	+0.0005 [-0.000, +0.001]	-0.016 [-0.016, -0.015]	-0.0095 [-0.010, -0.009]	0.44
ViT-g·2.0 surgery noDI head	+10.36 [+8.000, +12.767]	+0.119 [+0.115, +0.122]	-0.0077 [-0.025, +0.009]	-0.110 [-0.113, -0.108]						0.55
WINNER	S	P	P	P	P	P	P	P	P	